

Introduction to Non-Parametric Tests

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Most of the hypothesis testing procedures discussed so far, such as the *t-test* and *z-test*, are based on an important assumption.

- Random samples are selected from a normal population
- These tests allow some deviation from normality for large samples

Limitations of Parametric Tests

If the population distribution is:

- Badly skewed
- Grossly non-normal

For small samples:

- Parametric tests may not yield valid results

Assumptions of Parametric Tests

Most parametric tests assume that:

- Meaningful sample statistics can be computed
- Statistics such as mean and standard deviation are reliable

However:

- Nominal data (e.g., increase, decrease, no change)
- Ordinal or ranked data

do not satisfy these assumptions.

Non-Parametric (Distribution-Free) Tests

To handle such situations, statisticians developed alternative methods:

- Do not depend on the shape of the distribution
- Do not rely on population parameters (mean, variance)

These are called:

- Distribution-free tests
- Non-parametric tests

Historical Background

Important milestones:

- Chi-Square Test (Karl Pearson, 1900)
- Wilcoxon two-sample test (1945)

When Are Non-Parametric Tests Needed?

Non-parametric tests are useful when:

- Data are non-normal
- Distribution is bounded on one side
- Sample size is less than 50
- Population standard deviation is unknown
- Data are ranked or ordinal

Example of Non-Normal Distribution

- Income distributions are bounded at zero
- No practical upper limit
- Data tend to cluster near the lower bound

Non-Parametric Tests Covered

There are many non-parametric tests, but in this chapter we focus on:

- ① Sign Test
- ② Wilcoxon Signed Rank Test

Advantages of Non-Parametric Tests

Non-parametric tests offer several advantages, especially when the assumptions of parametric tests are not satisfied.

Distribution-Free Nature

- Non-parametric tests are distribution free
- No assumption is required about the population following:
 - Normal distribution
 - Any other specific distribution

Simplicity and Ease of Use

- Simple to understand
- Easy to apply
- Particularly useful when sample sizes are small

Computational Advantages

- Do not require lengthy or laborious computations
- Less time-consuming compared to parametric tests
- If significant results are obtained, no further analysis is required

Applicability to Different Types of Data

Non-parametric tests are applicable to:

- Qualitative data (nominal scale)
- Ranked data (ordinal scale)
- Precisely measured data:
 - Interval scale
 - Ratio scale

Use with Small Samples

- Suitable for very small sample sizes
- Useful for:
 - Pilot studies
 - Research involving rare diseases
 - Medical research with limited data

Fewer Assumptions

- Require fewer assumptions than parametric tests
- Assumptions are less stringent
- Assumptions are easier to satisfy in practical situations

The **Sign Test** is the simplest of the non-parametric tests.

- Based on the direction (sign) of differences
- Does not depend on numerical magnitude
- Uses only plus (+) and minus (−) signs

Basic Idea of Sign Test

In problems where the sign test is used, we count:

- Number of positive (+) signs
- Number of negative (−) signs
- Number of zero differences (discarded)

The test is based only on the signs of paired observations.

Null Hypothesis

The null hypothesis for the sign test is:

$$H_0 : p = 0.5$$

- If differences are due to chance:
- Probability of a + sign = $\frac{1}{2}$
- Probability of a - sign = $\frac{1}{2}$

- Let S be the number of times the less frequent sign occurs
- S follows a binomial distribution

$$S \sim \text{Binomial}(n, p = 0.5)$$

For a two-sided test at $\alpha = 0.05$, the critical value is:

$$K = \frac{n-1}{2} - 0.98\sqrt{n}$$

- Reject H_0 if:

$$S \leq K$$

Types of Sign Test

The sign test can be classified into:

- ① One-Sample Sign Test
- ② Paired-Sample Sign Test

One-Sample Sign Test

- Tests the hypothesis:

$$H_0 : \mu = \mu_0$$

- Replace each sample value:
 - Greater than μ_0 with (+)
 - Less than μ_0 with (-)
 - Equal to μ_0 are discarded
- Test whether signs follow a binomial distribution with $p = 0.5$

Paired-Sample Sign Test

Used in problems involving paired observations, such as:

- Collection of amount before and after a new policy
- Responses of mother and daughter towards ideal family size
- Assign:
 - (+) if first value
second value
 $>$
 - (−) if first value
second value
 $<$
 - Discard if equal
- Procedure is the same as the one-sample sign test

Problem Statement

A typing school claims that in a six week intensive course, it can train students to type on an average at least 60 words per minute. A random sample of 15 graduates is given a typing test and the median number of words per minute typed by each of these students is given below. Test the hypothesis that the median typing speed of graduates is at least 60 words. Test the hypothesis that the median typing speed of graduates is at least 60 words per minute using the Sign Test at 5% level of significance.

Observed Data

Student	WPM
A	81
B	76
C	53
D	71
E	66
F	59
G	88
H	73
I	80
J	66
K	58
L	70
M	60
N	56
O	55

Formulation of Hypotheses

$$H_0 : M_d = 60$$

$$H_1 : M_d > 60$$

Where M_d denotes the median typing speed of graduates.

Level of Significance

- Level of significance:

$$\alpha = 0.05$$

- Since the alternative hypothesis is one-sided, a one-tailed test is applied.

Observed Data

Student	WPM	WPM - 60	Sign
A	81	+21	+
B	76	+16	+
C	53	-7	-
D	71	+11	+
E	66	+6	+
F	59	-1	-
G	88	+28	+
H	73	+13	+
I	80	+20	+
J	66	+6	+
K	58	-2	-
L	70	+10	+
M	60	0	Discard
N	56	-4	-
O	55	-5	-

Counting of Signs

- Number of positive (+) signs = 9
- Number of negative (-) signs = 5
- Number of zero differences = 1 (discarded)

Effective sample size:

$$n = 15 - 1 = 14$$

- Let S be the number of times the less frequent sign occurs.
- Here:

$$S = 5$$

The critical value for the Sign Test is given by:

$$K = \frac{n-1}{2} - 0.98\sqrt{n}$$

Substituting $n = 14$:

$$K = \frac{14-1}{2} - 0.98\sqrt{14}$$

$$K = 6.5 - 3.67 = 2.83$$

Decision Rule

- Reject H_0 if:

$$S \leq K$$

- Here:

$$S = 5 \quad \text{and} \quad K = 2.83$$

Since:

$$S > K$$

- The null hypothesis is accepted.
- There is no sufficient evidence to conclude that the median typing speed exceeds 60 WPM.

Conclusion: The median typing speed of graduates is at least 60 words per minute.

Sign Test for Large Samples

- For large samples, generally $n > 25$, the normal approximation to the binomial distribution may be used.
- Continuity correction is applied.
- This approximation is valid since:

$$p_0 = 0.5$$

Mean and Standard Deviation

For the sign test with $p_0 = 0.5$:

$$\text{Mean} = np_0 = \frac{n}{2}$$

$$\text{Standard Deviation} = \sqrt{np_0(1 - p_0)} = \frac{1}{2}\sqrt{n}$$

These values are used to compute the Z statistic.

Test Statistic

The test statistic is given by:

$$Z = \frac{X - np_0}{\sqrt{np_0(1 - p_0)}}$$

Where:

- X = number of plus (+) signs
- n = effective sample size
- $p_0 = 0.5$

The computed value of Z is compared with the critical value from the standard normal table.

Handling of Ties

- Observations with zero difference are discarded.
- Effective sample size is reduced accordingly.
- Only positive and negative signs are used in computation.

Illustration : Problem Statement

The following data relate to the daily production of cement (in million tonnes) of a large plant for 30 days.

Test the null hypothesis that the plant's average daily production is 11.2 million tonnes against the alternative hypothesis:

$$H_1 : \mu < 11.2$$

at the 5% level of significance.

Formulation of Hypotheses

$$H_0 : \mu = 11.2 \text{ million tonnes}$$

$$H_1 : \mu < 11.2 \text{ million tonnes}$$

11.5, 10.0, 11.2, 10.0, 12.3, 11.1, 10.2, 9.6, 8.7, 9.3, 9.3,
10.7, 11.3, 10.4, 11.4, 12.3, 11.4, 10.2, 11.6, 9.5, 10.8,
11.9, 12.4, 9.6, 10.5, 11.6, 8.3, 9.3, 10.4, 11.5

This is a one-tailed (left-tailed) test.

Assignment of Signs

- Each observation is compared with 11.2
- A (+) sign is assigned if observation > 11.2
- A (-) sign is assigned if observation < 11.2
- Zero differences are discarded

Results obtained:

- Number of (+) signs = 11
- Number of (-) signs = 18
- Number of zeros = 1

Sample Size and Test Statistic

$$n = 30$$

$$X = 11$$

$$p_0 = \frac{1}{2}$$

Substituting in the Z-formula:

$$Z = \frac{11 - 30 \times \frac{1}{2}}{\sqrt{30 \times \frac{1}{2} \times \frac{1}{2}}}$$

Computation of Z

$$Z = \frac{11 - 15}{\sqrt{7.5}}$$

$$Z = \frac{-4}{2.74}$$

$$Z = -1.46$$

Decision Rule

- At $\alpha = 0.05$ (one-tailed test), the critical value is:

$$Z_{0.05} = -1.645$$

- Reject H_0 if $Z < -1.645$

Conclusion

$$Z = -1.46$$

Since:

$$Z < -1.645$$

- The null hypothesis is rejected.

Conclusion: The plant's average daily production of cement is less than 11.2 million tonnes.

Remarks on Sign Test

- Applicable to paired and independent samples
- Requires minimal information about population distribution
- Useful when only the direction of change is known

Although some statisticians object to its use for independent samples, the sign test remains valid and controls the probability of Type I error at the chosen level of significance.